

Practical Manual: The Live-Compute Enrichment Notebook

A self-paced walkthrough of
part2-synthetic-data-tutorial-practical-indabax-2026.Rmd | IndabaX Ghana
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How to use this manual

This deck walks you, at your own pace, through `part2-synthetic-data-tutorial-practical-indabax-2026.Rmd` — the notebook that runs the whole enrichment practical *inside itself* (no precomputed files).

- ▶ Open the notebook beside these slides.
- ▶ Each section here names the **notebook section**, tells you **what to run**, **what you will see**, and **what it means**.
- ▶ The numbers shown are illustrative — yours will be close.

Goal: use **GLSS6** (which observes welfare) to **impute** welfare for **DHS 2014** (which observes child health but not welfare), then study child outcomes by the imputed welfare/poverty status.

Before you start: setup

1. Install R and RStudio.
2. Install the packages:

```
install.packages(c("mice", "dplyr", "ggplot2", "tidyr",  
                  "rmarkdown", "kableExtra"))
```

3. Open
part2-synthetic-data-tutorial-practical-indabax-2026.Rmd.
Prefer a plain script? Open
part2-synthetic-data-tutorial-practical-script-indabax-2026.R.
4. Put the **real workshop extract** your facilitator shares in
data/practical/teaching/ (prepared teach_*.rds files — the notebook
never reads raw GLSS or DHS files). The WebR HTML file in webr/ uses
smaller hosted CSVs generated from the same extract.

The scale knobs (top of the notebook)

The model is computed live, so runtime is in your hands. Edit these YAML params.

Fast classroom demo

```
subsample_train: 3000      # GLSS rows used to fit
subsample_holdout: 1000    # GLSS holdout rows checked
subsample_dhs: 3000        # DHS rows enriched
m_implicates: 3            # imputed datasets
maxit: 2                   # chained-equation iterations
```

Full run

```
subsample_train: 0         # 0 means all rows
subsample_holdout: 0
subsample_dhs: 0
m_implicates: 5
maxit: 5
```

How to run it

- ▶ **All at once:** click **Knit** — produces the full HTML/PDF report.
- ▶ **Step by step:** run each code chunk with the green arrow, reading the output as you go.

The first time you run the *Run the Imputation* chunk, it fits the models — this is the slow step. Everything after it is instant because it just reads the results held in memory.

Section 1 — Dataset Warm-Up

Run: the load-data and warm-up chunks.

You will see: row/column counts, an “observed vs missing” map, side-by-side predictor tables, a welfare histogram, and a weighted-vs-unweighted comparison.

What it means:

- ▶ **GLSS** observes welfare; **DHS** observes child outcomes; **shared predictors** bridge them.
- ▶ Welfare is **right-skewed** → we model it on the log scale later.
- ▶ Weighted and unweighted rates differ (e.g., urban share) → **always use weights for rates.**

Section 2 — Run the Imputation (the heart of it)

Run: the run-imputation chunk. *This is the slow one.*

What it does, under the hood:

- ▶ Stacks **GLSS** training rows with the rows to complete, and **fits mice on the training rows only** (`ignore =`) — the model never learns from **DHS**.
- ▶ Models **log welfare** (`norm`) and **NHIS** (`logreg`), drawing `m` imputed datasets.
- ▶ Lets `mice` fill a few incidental missing cells in shared predictors, while the substantive target remains the missing welfare/NHIS/electricity block.
- ▶ **Derives** poverty status from each drawn welfare at GH¢792 / GH¢1,314 — never drawn directly.

What it means: this is “learn in **GLSS**, transport to **DHS**.” Everything after just displays its results.

Section 3 — Transporting imputed variables to DHS

Run: the enrichment chunks.

You will see: imputed welfare/poverty by imputed dataset, an NHIS imputed-vs-observed comparison, and a **derive-not-draw check** (should be **zero mismatches**).

What it means:

- ▶ **DHS** households now carry plausible welfare and poverty.
- ▶ Imputed NHIS reproduces the **GLSS level** (~7 pp below **DHS**) — the model faithfully transports the relationship, level gap and all.

Section 4 — Child stunting by imputed poverty

Run: the child-outcome chunks.

You will see: overall **DHS** stunting and child **stunting by the three imputed poverty status groups**, next to the **DHS wealth-index benchmark**.

What it means:

- ▶ The running question now has a consumption-based poverty gradient inside **DHS**.
- ▶ Collapsing very poor and poor to “poor” is still useful for the pooled gap, but the three-level table shows more of the poverty gradient.

Section 5 — Validation Check 1: GLSS holdout

Run: the holdout chunks. **You will see:** a validation table, a poverty-rate plot, a density overlay, a confusion matrix, and a subgroup table.

What good looks like:

- ▶ Imputed poverty rate near the true rate (e.g., ~18% vs ~16%), stable across imputed datasets.
- ▶ Imputed and true welfare **densities overlap**.
- ▶ Confusion matrix mass on the **diagonal**.

What it means: household-level correlation is modest by design (~0.5); the win is **rates, distributions, and subgroup patterns** — not one-to-one prediction.

Section 6 — Validation Check 2: transport

Run: the transport-check chunks.

You will see: predicted vs true **DHS electricity** rates, overall and by region/urban.

What it means:

- ▶ We hide a variable **DHS** does observe — a **validation target** — predict it from **GLSS**, and compare.
- ▶ A level gap (e.g., predicted ~73% vs true ~78%) is the **lesson**: a 2012/13 relationship does not perfectly transport to 2014.
- ▶ This checks transport for *observed* targets — **not welfare itself**. (The literature calls it *calibration*.)

Section 7 — Pooling across imputed datasets

Run: the rubin-style-pooling chunk.

You will see: the poor–non-poor stunting gap per imputed dataset, then a pooled estimate with a standard error.

What it means: one imputed dataset is never the answer. Average the estimate; the total variance adds **within**-imputation and **between**-imputation uncertainty (Rubin's rules).

Appendix — Predictor and target roles

Run only if you want the design audit: the manifest-roles, model-spec, and model-drops chunks near the end.

You will see: how many variables play each role, the model design table, and the three columns dropped from the model.

What it means:

- ▶ **Shared predictors** X , **synthetic targets** Y , derived poverty, and **child outcomes** each have a role.
- ▶ Three predictors (`members_total`, `adults_15plus`, `floor_non_earth`) are kept in the data but dropped from the model because they are exact combinations of other columns.

Appendix — Optional engine sensitivity

The `extension-engine-switch` chunk is visible in the appendix but set to `eval = FALSE`, so it does not run during the main practical. Run it manually, or change it to `eval = TRUE`, when you want to compare engines. Change one line:

```
engine <- "pmm"    # try "norm", "pmm", "cart", "rf"
```

You will see: a comparison of MAE, RMSE, correlation, and poverty rate against the norm baseline.

What it means: `cart` and `rf` are genuinely nonparametric/ML engines. They can tighten fit but may shift tails and subgroups — compare, don't assume.

Optional self-guided exercises

Use the appendix chunk when you want students to explore sensitivity after the main practical.

1. Change the welfare engine to "pmm", "cart", or "rf".
2. Remove one weak predictor and re-check holdout or transport performance.
3. Focus on one subgroup and decide whether the enrichment is credible enough for that policy question.

Email one group submission to augidenteh@gmail.com. The top two submissions will receive prizes.

Troubleshooting and tips

- ▶ **Too slow?** Lower the scale knobs (smaller subsamples, `m = 3`, `maxit = 2`).
- ▶ **Package errors?** Re-run the `install.packages(...)` line; restart R.
- ▶ **No data?** Unzip the workshop extract at the repo root so the files land in `data/practical/teaching/`.
- ▶ **Lost?** Each notebook section has a “How to read this” note right under its output.

Recap

You ran the full enrichment workflow yourself:

snapshot → holdout check → transport check → enrichment → policy gradient
→ pooling.

The discipline: be honest about what is observed, what is modeled, and what must be validated. The enrichment is useful when the checks line up — and the notebook makes every check visible.